1.) $N = 788_{10}$

a.) Division by 2 – BINARY

$788 \div 2$	394	0
$394 \div 2$	197	0
$197 \div 2$	98	1
$98 \div 2$	49	0
$49 \div 2$	24	1
$24 \div 2$	12	0
$12 \div 2$	6	0
$6 \div 2$	3	0
$3 \div 2$	1	1
$1 \div 2$	0	1

$= 1100010100_2$

b.) Division by 8 – OCTAL

$788 \div 8$	98	4
$98 \div 8$	12	2
$12 \div 8$	1	4
$1 \div 8$	0	1

$= 1424_8$

c.) Division by 16 – HEXADECIMAL

$788 \div 16$	49	4
$49 \div 16$	3	1
$3 \div 16$	0	3

$= 314_{16}$

2.) check by converting back to decimal with place-value table

Binary: 1100010100_2

2^9	2^8	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
512	256	128	64	32	16	8	4	2	1
1	1	0	0	0	1	0	1	0	0

$= 512 + 256 + 16 + 4 = 788$

Octal: 1424

8^3	8^2	8^1	8^0
512	64	8	1
1	4	2	4

$(1 \times 512) + (4 \times 64) + (2 \times 8) + (4 \times 1)$

$= 512 + 256 + 16 + 4 = 788$

Ans

Hexadecimal: 314_{16}

16^2	16^1	16^0
256	16	1
3	1	4

$= (256 \times 3) + (16 \times 1) + (1 \times 4)$

$= 768 + 16 + 4 = 788$

3.) Convert Binary to octal and Hex by grouping

Binary: 1100010100_2

To Octal; Group from 3 from right

001 100 010 100

$= 1 \quad 4 \quad 2 \quad 4$

$= 1424_8$

To Hex: Group by 4 from right

0011 0001 0100

$= 3 \quad 1 \quad 4$

$= 314_{16}$

Error Analysis: The mistake is that my classmate missed one bit and reverse the remainder order, so instead of 101110_2 which equals 46_{10}, they write 011101_2 which only equals 29_{10}.